

# COMP9517: Computer Vision

## 2025 T3 Lab 4 Specification

### Maximum Marks Achievable: 2.5

This lab is **worth 2.5% of the total course marks.**

The lab files must be submitted online.

Instructions for submission will be posted closer to the deadline.

**Deadline for submission is Week 7, Monday 27 October 2025, 18:00:00 AET.**

**Objective:** This lab revisits important concepts covered in the Week 5 lectures and aims to make you familiar with implementing specific algorithms.

**Software:** You are required to use OpenCV 3+ with Python 3+ and submit your code as a Jupyter notebook (see coding and submission requirements below). In the tutor consultation session this week and next, you can ask any questions you may have about this lab.

**Materials:** The sample images to be used in this lab are available via WebCMS3.

**Submission:** All code and requested results are assessable after the lab. Submit your source code as a Jupyter notebook (.ipynb file) that includes all outputs (see coding requirements at the end) by the above deadline. The submission link will be announced in due time.

#### Background

In many computer vision applications, image segmentation is a crucial first step towards quantitative image analysis. As discussed in the lectures, there are many possible image segmentation methods, ranging from very simple to very complicated.

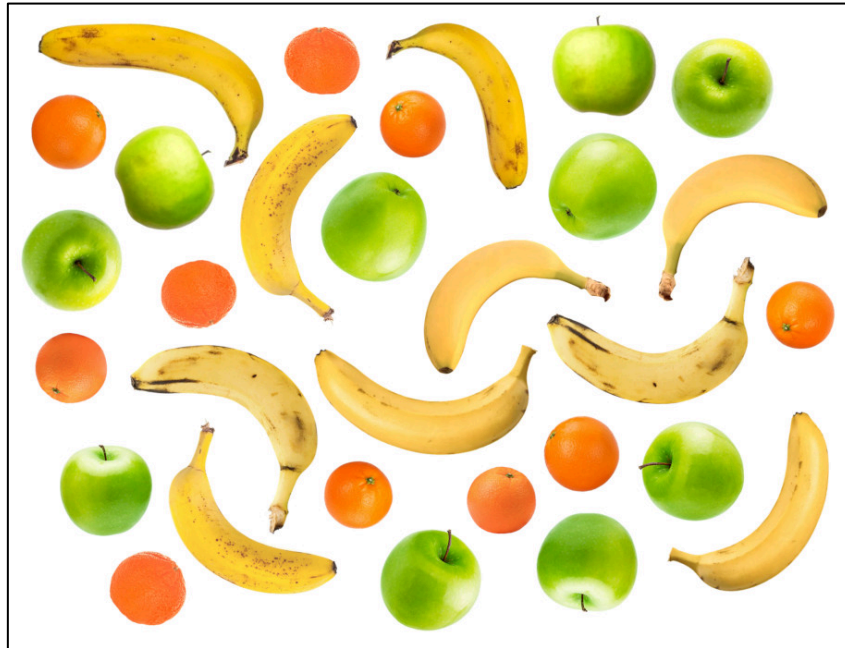
Despite its simplicity and limitations, **intensity thresholding** is still a very popular image segmentation method. This is because imperfections in the segmentation results can often be filtered out by postprocessing using binary morphological operators.

The goal of this lab is to **write an algorithm that performs image segmentation and subsequent analysis based on just intensity thresholding, binary morphological processing, and applying some basic postprocessing rules to the segmented objects.**

 The scenario is as follows. A company in the food industry wants to develop a quality control pipeline that fully automatically checks whether each batch of fruit contains the right number

labels

of apples, bananas, and mandarins before packaging. Each batch must contain exactly  $N$  of each type of fruit, where  $N$  is a number set by the human operator of the pipeline. In our case,  $N = 10$ . Each batch is imaged (see below for an example) and checked automatically.



**#Apples = 10, #Bananas = 10, #Mandarins = 10, All good**

### Task (2.5 marks)

You are responsible for developing the algorithm that performs the task. More specifically, for each batch image, the algorithm must count the number of apples, bananas, and mandarins, report these numbers, and indicate whether the batch needs correction and how.

For example, if the count for each type of fruit is  $N$ , no further action is needed, and the algorithm must indicate "All good". But if the count for any type is less than  $N$ , the algorithm must indicate "Add  $X$  apples" and/or "Add  $Y$  bananas" and/or "Add  $Z$  mandarins" (whichever is applicable), where  $X$ ,  $Y$ , and  $Z$  are the numbers needed to get to  $N$ . Conversely, if the count is larger than  $N$ , the algorithm must indicate "Remove  $X$  apples" and/or "Remove  $Y$  bananas" and/or "Remove  $Z$  mandarins" (whichever is applicable) to get to  $N$ .

In your code, specify  $N$  at the beginning and then use it throughout the algorithm, so that it can be easily set by the user as needed. Also, if you use any operations with (hyper)parameters, define their values at the beginning of your code, too.

Your algorithm must work fully automatically without any user input or interaction during operation. It must be able to handle all images provided in this lab (see WebCMS3) with the same (hyper)parameter values (not different values per individual image).

Hints: Convert the input image to grayscale (if needed), apply intensity thresholding (using either an automatically calculated or a manually fixed threshold), use binary morphology operators to fill any holes or filter out noise pixels in the segmentation, run the connected components algorithm or something similar to identify and count the separate regions, compute shape or size or colour features (or any combination of them) to determine whether a region is an apple, banana, or mandarin, and implement some rules to calculate and print out what action is needed (as described above).

### **Coding Requirements**

Make sure that in your Jupyter notebook, the input images are readable from the location specified as an argument, and all results are displayed in the notebook environment. All cells in your notebook should have been executed so that the tutor/marker does not have to execute the notebook again to see the results.

If your algorithm has any (hyper)parameters, their values must be specified at the beginning of your code, so that users can easily change and experiment with these parameters.

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**Released:** 13 October 2025